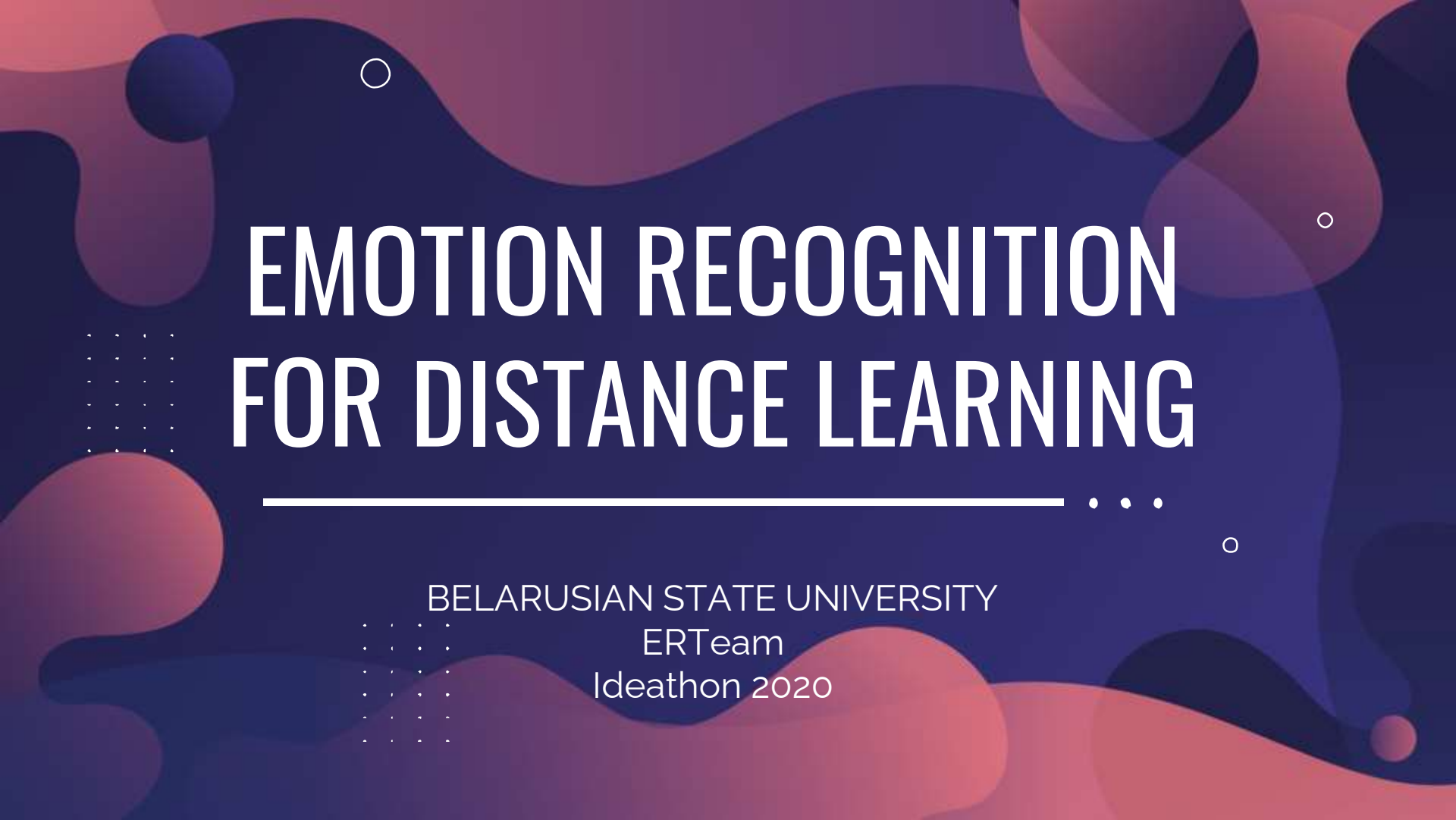




EMOTION RECOGNITION FOR DISTANCE LEARNING

BELARUSIAN STATE UNIVERSITY

ERTeam

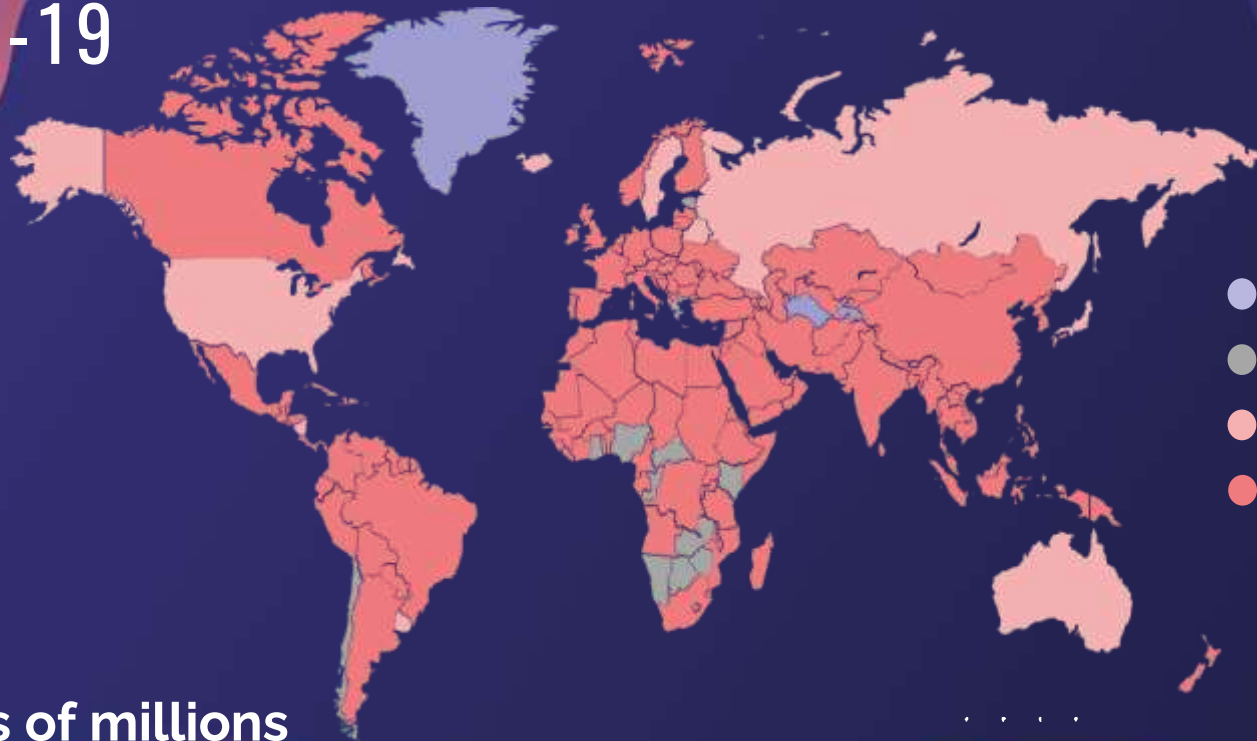
Ideathon 2020

Web – application

to recognize emotions through facial expressions in video conferences



COVID-19



Hundreds of millions
of students are in need of distance
learning because of the coronavirus situation



More 90,37 billions

However, the outbreak of **COVID-19** has paved opportunities for the growth of the distance learning segment

Name: ERSchool



SERVICE: foreign language teaching

FORMAT: distance learning through video conferences with groups of 5-10 persons

CUSTOMERS: adults and teenagers

TIME ON MARKET: 15 years



GOALS:

- High level of service
- Don't lose clients



VALUES:

- Comfort state of employees and students and they enjoy using service



PRIVATE POLICE:

- adhere the private police of each customer
- data processing with customer consent



FEARS:

- provide low level service
- reduce loyalty of clients

CURRENT SITUATION OF ERSCHOOL



Low student's engaged

Only 10 % of students according to research continue active education throughout the course



Burnout of professors

As a consequence, low level of productivity and increase rates of resignation



High drop-out students

Drop-out rates for distance learning students are higher than traditional

TRIGGERS



Burnout is the reason of
EMPLOYEE DISCHARGE



NEW CLIENT in 7 times expensive
than keeping established
customer



SOLUTION

RECOGNIZE EMOTIONS WITH A NEURAL NETWORK



predict emotional
burnout of employees

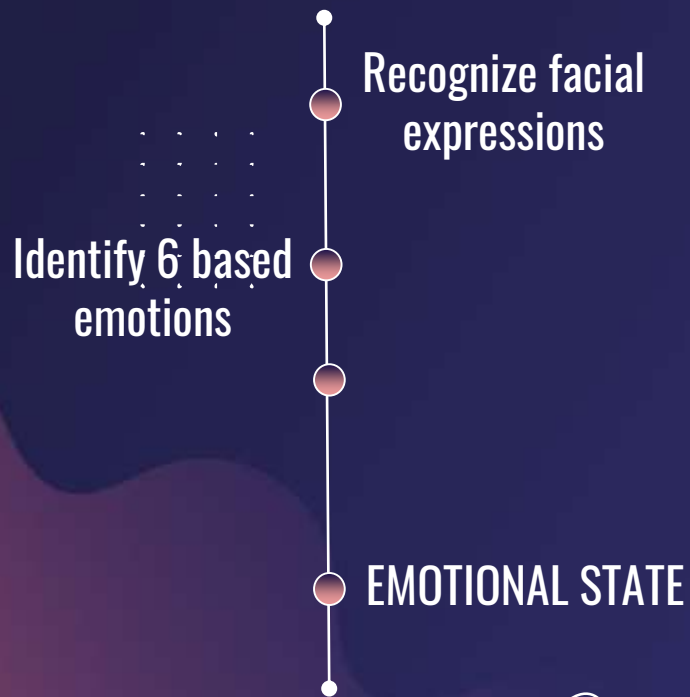


student's and
teacher's emotional
statistic



predict refusing from
educational serves

SCIENTIFIC RESEARCH



angry



happy



surprise



disgust



sad



fear

WITHOUT OUR APPLICATION

TEACHER IS AT RISK
OF EMOTIONAL BURN

Low productivity and
negative emotion state

Low satisfaction
factor

DISCHARGE

USE OUR APPLICATION

TEACHER IS AT RISK
OF EMOTIONAL BURN

Company takes
actions

Increase productivity of
educational process

HIGH PROFIT

We predict it

WITHOUT OUR APPLICATION

LOW STUDENT'S ENGAGED

Low productivity and
negative emotion
state

Low satisfaction
factor

REFUSAL

USE OUR APPLICATION

LOW STUDENT'S ENGAGED

Company
takes actions

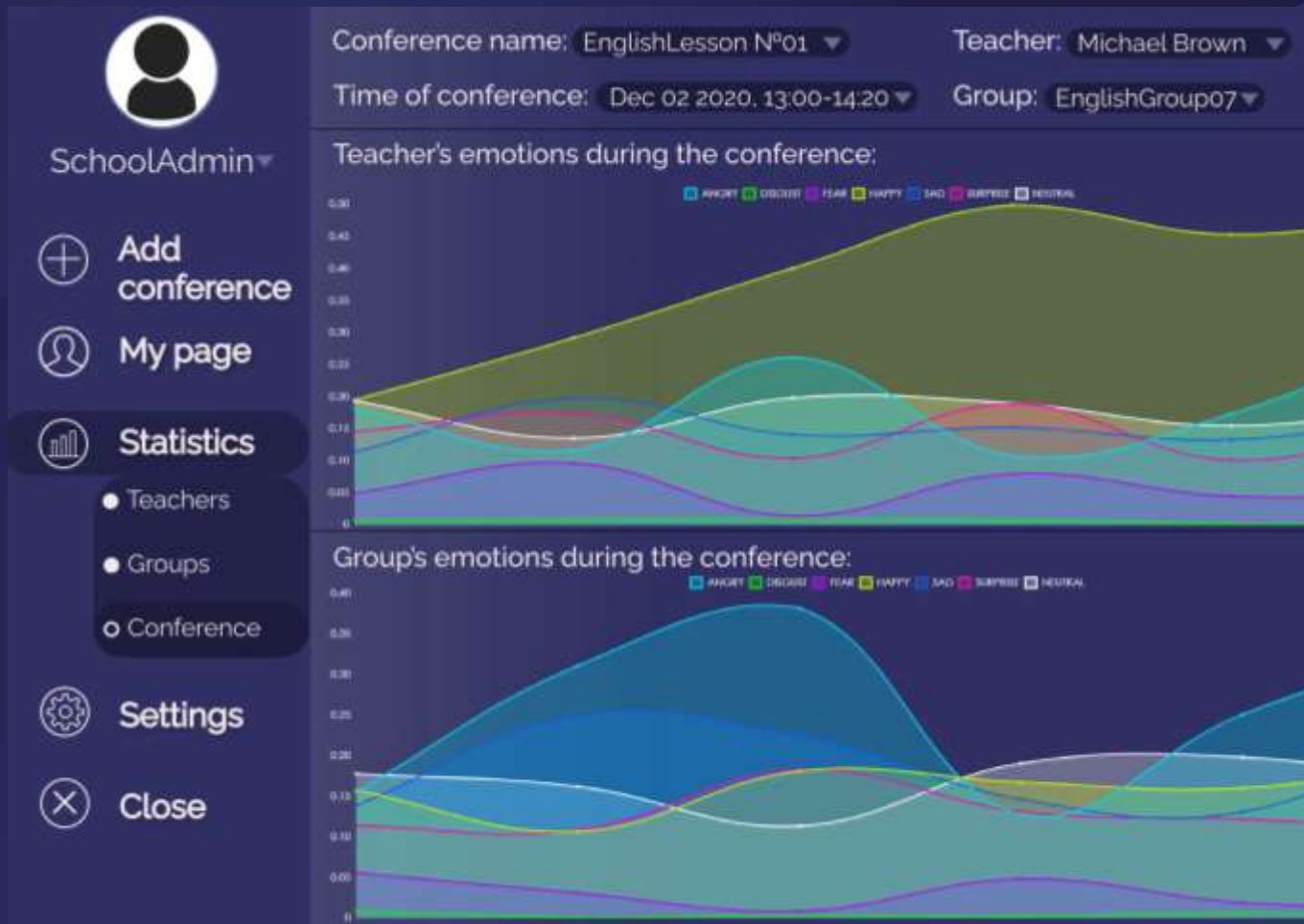
Increase
productivity of
educational process

HIGH PROFIT

We predict
large
number of
drop-out
students

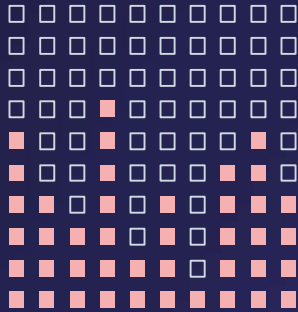
PROTOTYPE

on the stage
collecting
statistics



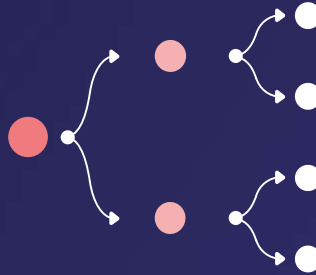
PROTOTYPE

PREDICT REFUSING FROM EDUCATIONAL SERVICE



STEP 1

Collect top student's emotions during class work



STEP 2

Learning neural network using drop-out statistics of company

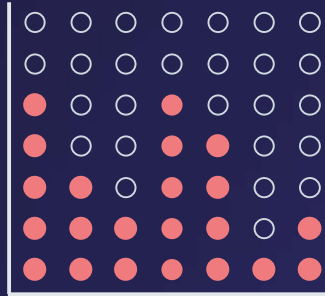


STEP 3

Predict refusing from educational serves

PROTOTYPE

PREDICT EMOTIONAL BURNOUT OF EMPLOYEES



STEP 1



STEP 2



STEP 3

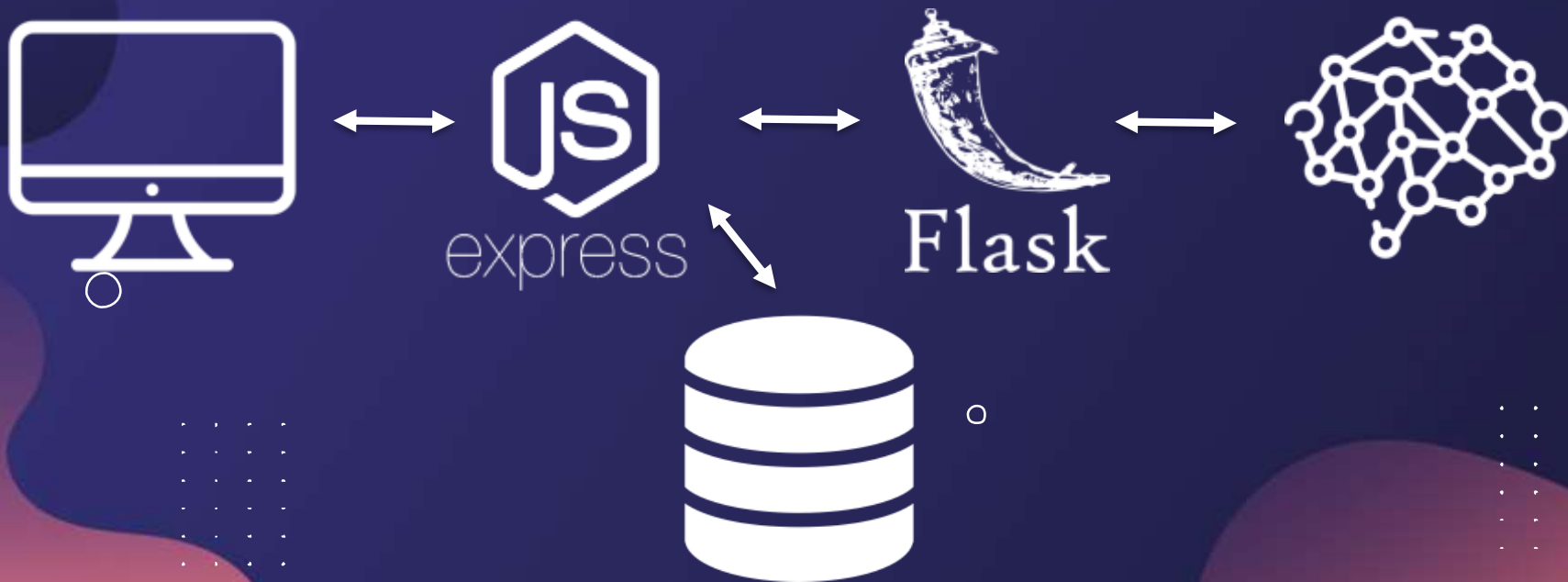
Collect top
teacher's emotions
during class work

Learning neural network
using burnout statistics of
company

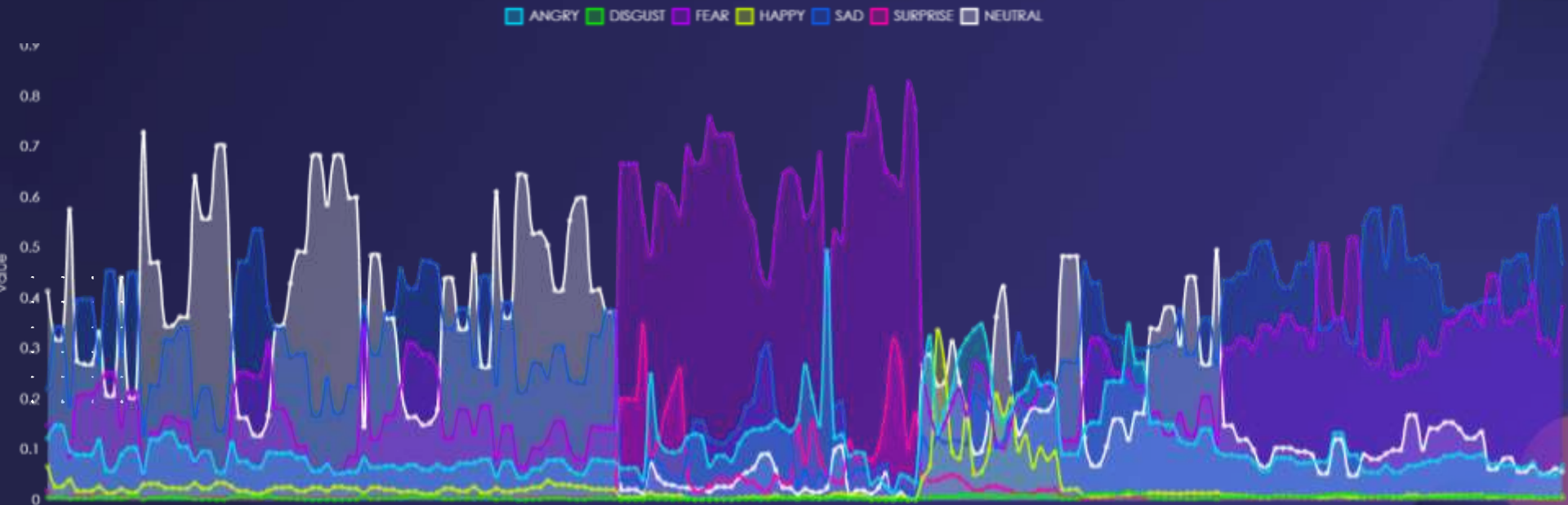
Predict emotional
burnout of employees

ARCHITECTURE

: of application



TEST on OUR TEAM CONFERENCE



DIRECTION OF DEVELOPMENT OF OUR PROJECT

WE HAVE:

STUDY SCIENTIFIC RESEARCHES
CONCERNING EMOTIONS
RECOGNITION AND INFLUENCE
EMOTIONS IN EDUCATION

PROTOTYPE OF CLIENT AND SERVER
PARTS OF APPLICATION

EMOTION RECOGNITION BASIC
EMOTIONS FOR GROUP



DATABASE

Create dataset for learning additional
Neural Network on this emotions

VOICE



Considerable part of emotion state can
be identified by voice

NEURAL NETWORK



Developing additional neural
networks

○



 Developer
@klklklklklklklklklklklkl



 Data scientist
@yana_CHERNYAK



Developer
@ArtemOpanasenko



Mentor



THANKS!

Do you have any questions?

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